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**EE-7101-GS**  
**B.E. VII Semester**  
Examination, December 2020  
**Grading System (GS)**  
**Computer Aided Design of Electrical Machines**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. What do you understand by the phenomena of cogging and crawling in an induction motor? What steps will you suggest at the design stage to avoid this?
2. a) What is meant by SCR of an alternator? Discuss its significance in relation to stability voltage regulation and parallel operation of synchronous generators.  
b) What are the designs steps followed for designing motor and rotor winding of a turbo alternator.
3. What do you understand by short circuit ratio of an alternator? How it affects the performance of the machine? Explain.
4. Design a 8 section starter for a 20hp, 250V, 1000rpm DC shunt motor from the following data:  
Max. torque = F.L. torque, Arm. resistance =  $0.4\Omega$ ,  
Motor efficiency = 85%,  
Also find the speeds at which notching takes place.

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5. Draw the equivalent circuit and design the single phase induction using Genetic Algorithm.
6. Explain clearly the various steps involved in design of a wound rotor of a 3 phase induction motor and indicate how do you estimate the various losses in an induction motor from design data.
7.
  - a) Explain how the starting torque of an induction motor can be improved by changing the rotor design.
  - b) Derive the output equation of a 3-phase induction motor in terms of machine design parameters.
8. Write a short notes (Any two)
  - i) Design of slip ring and squirrel cage rotor
  - ii) Design consideration of turbo and water wheel alternators.
  - iii) Optimization of design constrained and unconstrained optimization problem.
  - iv) Design of field systems of synchronous machine

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